GitHub Repository: https://github.com/Khushaligupta01/ShortlistAI

ShortlistAI — Technical Design Document (TDD)

System Name: ShortlistAI

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Authors: Principal Software Architect, Senior DevOps Engineer, AI Governance Engineer, Full-Stack Development Team

Status: Approved for Release & Take-Home Evaluation

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Executive Summary

ShortlistAI is an enterprise-grade, evidence-grounded candidate screening and client shortlist delivery platform engineered for staffing agencies and recruitment consultancies. Modern hiring pipelines are overwhelmed by high-volume applicant flows, while conventional Applicant Tracking Systems (ATS) rely either on naive keyword matching (which rejects qualified talent over minor vocabulary discrepancies) or uncalibrated generative AI (which hallucinates qualifications, produces subjective scores, and introduces algorithmic bias).

ShortlistAI solves this problem through an architectural **Separation of Cognitive Concerns**:

- **AI as an Information Extractor (Never a Judge):** Large Language Models (LLMs) are strictly constrained via structured JSON schemas to locate and cite verbatim quotes from resumes that demonstrate specific rubric criteria. The LLM is explicitly barred from generating numerical scores or subjective hiring recommendations.
- **Deterministic Mathematical Scoring Engine:** A transparent, rule-based Python evaluation engine validates evidence, applies recruiter-approved weights across four criteria pillars, enforces hard knockouts, and calculates a normalized 0–100% match score.

With multi-tenant role isolation, privacy-preserving blind client portals, end-to-end GDPR consent governance (Article 17 erasure), and immutable audit logging, ShortlistAI delivers transparency, accountability, and defensibility for high-stakes hiring workflows.

1. Project Overview

1.1 Problem Statement

Recruitment agencies and corporate talent acquisition teams spend up to 75% of their working hours manually triaging resumes. Existing software solutions suffer from critical flaws:

1. **Keyword ATS Filters:** Reject qualified candidates due to trivial vocabulary discrepancies (e.g., "PostgreSQL" vs. "Relational SQL") and are easily manipulated by "white-font" keyword stuffing.
2. **Generative AI Scoring (Black Box):** Asking Large Language Models to assign subjective grades (e.g., "Rate this resume 1 to 10") leads to high variance, fabricated qualifications (hallucinations), and severe algorithmic bias against non-traditional backgrounds.
3. **Disconnected Recruiter-Client Handoffs:** Sending client hiring managers unformatted resumes via email attachments causes lost feedback, data leaks, and compliance violations under GDPR and the EU AI Act.

1.2 Target Personas

1. **Recruiter / Talent Sourcer:** Ingests bulk resumes (up to 500 CVs per batch), generates and fine-tunes job scorecards, reviews AI-extracted evidence citations, manages candidates through an interactive Kanban pipeline, and submits curated shortlists to clients.
2. **Delivery Head / Account Director:** Oversees cross-requisition pipeline velocity, monitors agency-wide candidate placement, audits scorecard rubric fairness, and enforces operational SLAs.
3. **Client Hiring Manager:** Accesses a secure, dedicated client portal to review blind-screened (PII-masked) candidate submissions, provides one-click feedback (Accept, Hold, Reject), and requests interviews.
4. **Candidate:** Maintains sovereignty over personal data, grants or withdraws consent via self-service verification tokens, and exercises the GDPR Right to be Forgotten.

1.3 Key System Capabilities

- **Automated Scorecard Rubric Generation:** Transforms raw job descriptions into 4-pillar structured rubrics (Required Skills, Preferred Skills, Experience, Education) with configurable weights and hard knockouts.
- **Dual-Engine Evaluation Architecture:** Combines LLM evidence extraction with a deterministic mathematical scoring engine.
- **High-Throughput Asynchronous Processing:** Parses PDF and DOCX files in background workers without blocking HTTP request threads, with tested capacity for batches up to 500 CVs.
- **Multi-Tenant Organization Isolation:** Enforces strict logical tenancy across staffing agencies and client accounts.
- **Privacy-Preserving Client Portal:** Features blind candidate masking to mitigate unconscious bias and protect candidate contact details.
- **Comprehensive Audit Trail & Consent Lifecycle:** Tracks consent states (`Granted`, `Pending`, `Withdrawn`) and logs immutable events with SHA-256 hashes.

2. Problem Analysis & Market Gap

Evaluation Dimension	Legacy Keyword ATS	Uncalibrated Generative AI	ShortlistAI (Our Solution)
Scoring Mechanism	Naive string matching / regex	Generative token likelihood ("Score: 8/10")	**Deterministic Weighted Math**
Explainability	Zero (Binary keyword hit/miss)	Hallucinated justifications	**Verbatim Evidence Citations**
Missing Information	Penalized as negative	LLM guesses or assumes	**Neutral zero score (No assumptions)**
Knockout Handling	Flaky regex filters	Unreliable prompting	**Strict Deterministic Guardrails**

Client Collaboration	Email attachments & PDFs	Ad-hoc spreadsheet exports	**Secure Multi-Tenant Client Portal**
Compliance & Consent	Unmonitored data storage	Cloud vendor data leakage risks	**GDPR Art. 17 Deletion & Full Audit**

3. Project Scope & Core Assumptions

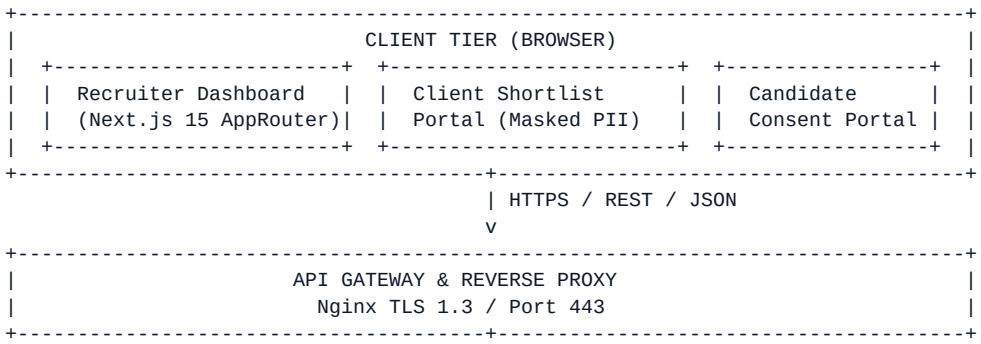
3.1 In-Scope Deliverables

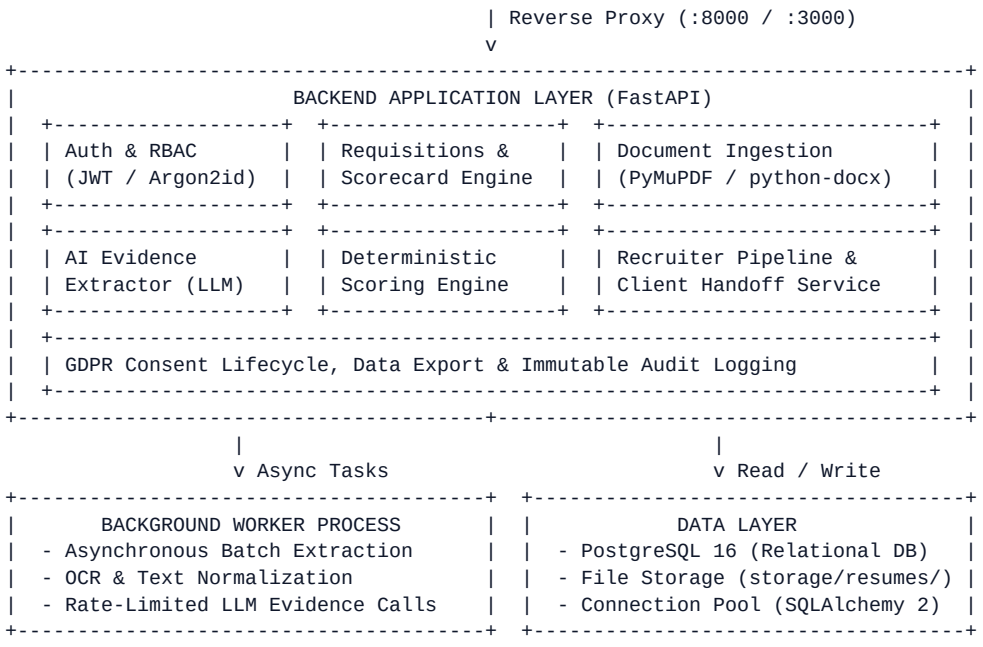
- Multi-role web application with dedicated views for Recruiters, Delivery Heads, and Client Hiring Managers.
- Scorecard creation from raw job descriptions with editable weights and knockout toggles.
- Batch resume ingestion supporting PDF and DOCX files, duplicate hash detection, and batch size boundary enforcement (up to 500 CVs).
- Evidence-backed AI extraction using OpenAI GPT-4o-mini with fallback to offline mock provider.
- Deterministic 4-pillar scoring engine with knockout enforcement.
- Candidate search, multi-faceted filtering, detail inspector, and Kanban pipeline.
- Client submission workflow with automatic contact PII masking.
- GDPR privacy suite: consent status tracking, public token-based candidate verification, and Article 17 hard deletion.
- Append-only audit logging for all critical system actions.

3.2 Engineering Assumptions

1. **Single Agency Multi-Tenant Model:** The primary deployment model serves one recruitment agency managing multiple client organizations with strict data isolation.
2. **Resume File Integrity:** Resumes are submitted as text-based PDFs or DOCX files. Image-only scanned PDFs are extracted where OCR engines are available, but text-layer PDFs provide optimal extraction fidelity.
3. **LLM Availability:** The platform defaults to OpenAI `gpt-4o-mini` for production workloads and provides a deterministic, zero-dependency `MockAIProvider` for offline development, automated testing, and CI/CD pipelines.
4. **Document Storage:** Files are stored on local persistent disk volumes (`storage/`) structured by organization UUID hashes, with architecture ready for S3/blob storage mounting.

4. System Architecture





4.1 Data Flow: Requisition & Scorecard Generation

1. Recruiter inputs Job Title, Department, and raw Job Description.
2. FastAPI persists the Requisition in `DRAFT` status.
3. AI Scorecard Extractor parses the description into structured criteria across 4 pillars.
4. Criteria weights are normalized to sum to exactly 100%.
5. Recruiter reviews, adjusts weights/knockouts, and approves the scorecard (`status = APPROVED`).

4.2 Data Flow: Batch Ingestion & Deterministic Ranking

1. Recruiter uploads a batch of resume files (PDF/DOCX, validated up to 500 files).
2. Files are hashed (SHA-256) for duplicate detection and saved to `storage/resumes/{org_id}/`.
3. Background worker extracts plain text using PyMuPDF / python-docx.
4. AI Evidence Extractor scans the extracted text against approved scorecard criteria, returning verbatim quotes and confidence scores.
5. Deterministic Scoring Engine computes weighted scores, evaluates knockouts, assigns fit tiers (`HIGH`, `MEDIUM`, `LOW`), and stores results.
6. Recruiter views ranked leaderboard with evidence citations and criteria breakdown.

5. Technology Choices & Trade-Off Analysis

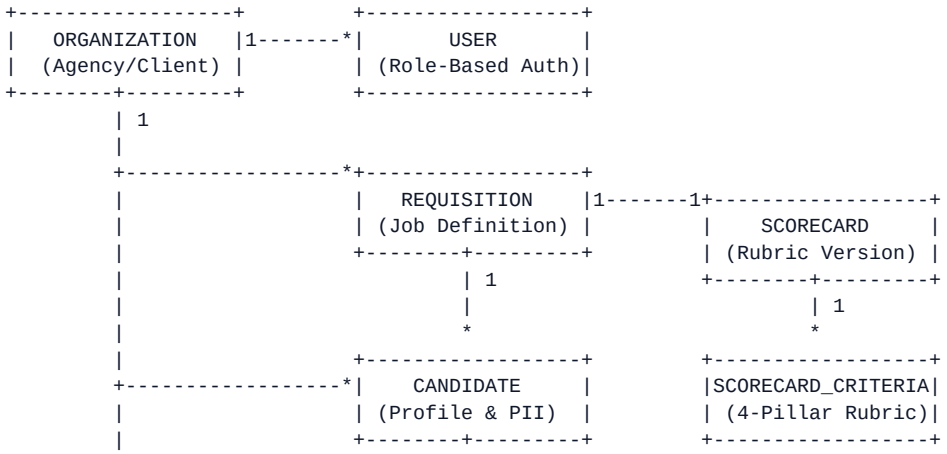
Component	Selected Technology	Alternative Evaluated	Trade-Off Rationale

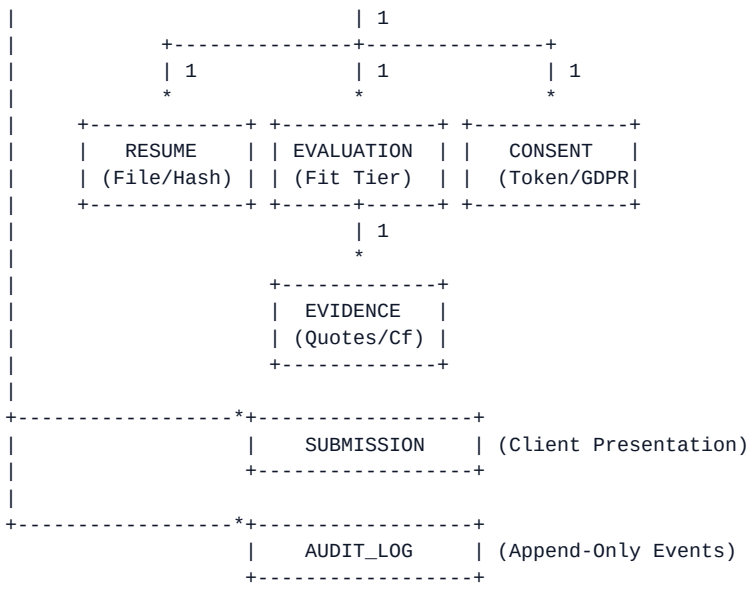
Backend Framework	**FastAPI (Python 3.10+)**	Django Ninja, Flask, Go Gin	High-concurrency async/await, native Pydantic v2 validation, automatic OpenAPI documentation. Python is mandatory for native NLP and document processing libraries.
Database ORM	**SQLAlchemy 2.0 (Async)**	Django ORM, Tortoise ORM, Prisma	Enterprise-grade query composition, robust connection pooling, explicit transactional boundaries, and native async support via `asyncpg`.
Frontend Framework	**Next.js 15 (App Router)**	Vite + React SPA, Remix	React Server Components, server-side data fetching, automatic route-level code splitting, standalone compilation (< 120MB container footprint).
Styling & Tokens	**Tailwind CSS + Lucide Icons**	Material UI, Ant Design	Zero-runtime CSS, responsive utility architecture, easily maintainable design tokens for the premium warm-neutral palette.
PDF Extraction	**PyMuPDF (fitz) + python-docx**	Tesseract, PyPDF2, PDFMiner	PyMuPDF executes 10x faster than PDFMiner with superior multi-column layout fidelity, native C bindings, and low memory overhead.
AI Provider Abstraction	**Pluggable Multi-Provider (OpenAI / Mock)**	LangChain, LlamaIndex	Custom lightweight provider avoids bloated dependencies, isolates prompt contracts cleanly, and allows instant offline mocking for tests and CI/CD.

6. Database Design & Multi-Tenant Data Architecture

The data tier is built on **PostgreSQL 16** with strict relational integrity, foreign key cascading constraints, and logical multi-tenant organization scoping.

6.1 Core Entities and Relationships





6.2 Key Entity Attributes

- **Organization:** `id (UUID PK)`, `name (VARCHAR)`, `org_type (ENUM: AGENCY, CLIENT)`, `created_at`.
- **User:** `id (UUID PK)`, `organization_id (UUID FK)`, `email (VARCHAR UK)`, `hashed_password`, `role (ENUM: RECRUITER, DELIVERY_HEAD, CLIENT_HIRING_MANAGER, SUPER_ADMIN)`, `is_active`.
- **Requisition:** `id (UUID PK)`, `organization_id (UUID FK)`, `title`, `department`, `status (ENUM: DRAFT, ACTIVE, CLOSED)`, `job_description`.
- **Scorecard:** `id (UUID PK)`, `requisition_id (UUID FK)`, `version (INT)`, `status (ENUM: DRAFT, APPROVED)`.
- **ScorecardCriteria:** `id (UUID PK)`, `scorecard_id (UUID FK)`, `category (ENUM: REQUIRED_SKILL, PREFERRED_SKILL, EXPERIENCE, EDUCATION)`, `name`, `weight (FLOAT)`, `is_knockout (BOOL)`.
- **Candidate:** `id (UUID PK)`, `organization_id (UUID FK)`, `requisition_id (UUID FK)`, `full_name`, `email`, `phone`, `status (ENUM: UNREVIEWED, SHORTLIST, HOLD, REJECT)`.
- **CandidateEvaluation:** `id (UUID PK)`, `candidate_id (UUID FK)`, `scorecard_id (UUID FK)`, `overall_score (FLOAT)`, `fit_tier (ENUM: HIGH, MEDIUM, LOW)`, `knockout_triggered (BOOL)`.
- **EvaluationEvidence:** `id (UUID PK)`, `evaluation_id (UUID FK)`, `criterion_id (UUID FK)`, `verbatim_quote (TEXT)`, `match_confidence (FLOAT)`.
- **Submission:** `id (UUID PK)`, `candidate_id (UUID FK)`, `client_organization_id (UUID FK)`, `status (ENUM: SUBMITTED, INTERVIEW_REQUESTED, REJECTED)`, `recruiter_notes`, `client_feedback`.
- **ConsentRecord:** `id (UUID PK)`, `candidate_id (UUID FK)`, `status (ENUM: GRANTED, PENDING, WITHDRAWN)`, `verification_token (VARCHAR UK)`, `consent_date`.
- **AuditLog:** `id (UUID PK)`, `organization_id (UUID FK)`, `user_id (UUID FK)`, `action`, `entity_type`, `entity_id`, `ip_address`, `timestamp`.

6.3 Multi-Tenant Isolation Strategy

- Every business table enforces an `organization_id` foreign key.
- Route dependencies inject the current authenticated user's `organization_id`.

- Client portal submissions are bridged through explicit `Submission` junction records. Client users can **only** query candidates submitted to their specific client organization ID; direct candidate tables remain inaccessible.

7. Authentication, Authorization & Role Permissions

ShortlistAI uses JWT access tokens signed with HMAC-SHA256 and passwords hashed with Argon2id / bcrypt.

7.1 Role-Based Access Control (RBAC) Matrix

Resource / Action	Recruiter	Client Hiring Manager	Delivery Head	Super Admin / Compliance
Create Requisition & Scorecard	Yes	No	Yes	Yes
Upload Resumes & Ingest Batches	Yes	No	Yes	Yes
Trigger AI Evaluation & Scoring	Yes	No	Yes	Yes
View Full Candidate PII (Email/Phone)	Yes	No (Masked)	Yes	Yes
Update Pipeline Stage (Shortlist/Reject)	Yes	No	Yes	Yes
Submit Candidate to Client Portal	Yes	No	Yes	Yes
Review Masked Client Submissions	Yes	Yes	Yes	Yes
Request Interview / Client Decision	No	Yes	Yes	Yes
View Agency-Wide Analytics & Metrics	No	No	Yes	Yes
Access GDPR Privacy & Deletion Center	No	No	No	Yes
View Immutable Audit Logs	No	No	Yes	Yes

8. Main Workflows

8.1 Workflow A: Job Description to Approved Scorecard

1. Recruiter navigates to `/requisitions/new` and enters job title, department, and full text description.
2. System calls `/api/v1/requisitions` to create the requisition record.
3. AI Scorecard Extractor analyzes the text and proposes structured criteria across 4 categories.
4. Recruiter reviews criteria on the interactive scorecard editor, modifies weights (validated to sum to 100%), toggles knockout flags, and clicks **Approve Scorecard**.

8.2 Workflow B: Batch Resume Ingestion & Processing

1. Recruiter uploads resumes (drag-and-drop PDF/DOCX files, up to 500 per batch) at `/requisitions/[id]`.
2. Backend validates file extensions, MIME types, and file size limits (15 MB max per file).
3. Backend performs SHA-256 content hashing to detect duplicate uploads across candidates.
4. Resumes are stored on disk under `storage/resumes/{org_id}/` with UUID filenames.
5. Background worker extracts plain text and creates initial Candidate records in `UNREVIEWED` status.

8.3 Workflow C: Evidence Extraction & Deterministic Evaluation

1. Worker triggers evidence extraction for each candidate against active scorecard criteria.
2. AI extracts verbatim quotes proving or disproving qualifications.
3. Scoring Engine calculates the weighted match score and flags any violated knockout criteria.
4. Evaluation record is stored with fit tier (`HIGH`: >=80%, `MEDIUM`: 60-79%, `LOW`: <60%).
5. Recruiter views ranked leaderboard ordered by match score.

8.4 Workflow D: Recruiter Review & Client Handoff

1. Recruiter inspects candidate profile: reviews side-by-side resume text and evidence citations.
2. Recruiter transitions qualified candidates to `SHORTLIST` with private recruiter notes.
3. Recruiter clicks **Submit to Client**, selecting the target client organization.
4. System creates a `Submission` record with contact information masked (e.g., `J* D`, `c***@example.com`).

8.5 Workflow E: Client Portal Decision Loop

1. Client Hiring Manager signs into `/client`.
2. Client views submissions assigned to their organization, reviewing blind candidate summaries.
3. Client clicks **Request Interview**, **Hold**, or **Decline**, providing qualitative feedback.
4. Decision updates the submission status and is instantly reflected on the Recruiter's Kanban board.

9. Resume Parsing & Ingestion Architecture

9.1 Multi-Format Document Processing

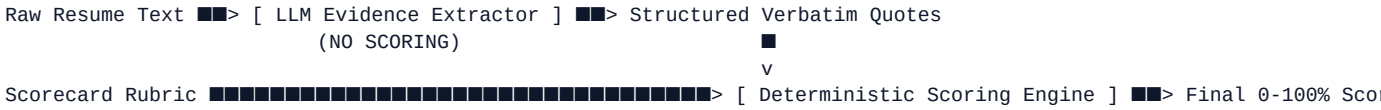
- **PDF Documents:** Processed using **PyMuPDF (`fitz`)**, which extracts textual streams while preserving layout blocks, multi-column formats, and structural reading order.
- **DOCX Documents:** Processed using `python-docx`, traversing paragraph blocks, bullet lists, and embedded table cells.
- **Normalization:** Extracted text is normalized by stripping null bytes, normalizing unicode ligatures, removing excessive whitespace, and redacting credit card/SSN patterns.

9.2 Batch Ingestion Capacity (Up to 500 Resumes)

- The system includes explicit boundary validation: batches exceeding 500 files are rejected with HTTP 400 Bad Request to protect system memory and file handle descriptors.
- Ingestion runs asynchronously: files are acknowledged immediately (HTTP 202 Accepted) with a `batch_id`, while background worker processes extract text and trigger evaluation in worker threads.
- Duplicate detection compares SHA-256 hashes of incoming files against existing stored resumes within the organization.

10. AI Extraction & Evaluation Approach

10.1 Separation of Cognitive Concerns



10.2 Structured JSON Schema Output

The AI prompt instructs the model to return strictly structured JSON matching a Pydantic schema:

```
{
  "criteria_evidence": [
    {
      "criterion_id": "uuid-here",
      "found": true,
      "verbatim_quote": "5 years of experience architecting distributed backend services in Python and FastAPI",
      "confidence": 0.95
    }
  ]
}
```

10.3 Failure Modes and Fallback Protection

- **Malformed JSON:** The backend catches JSON decode errors, re-attempts extraction with exponential backoff, and falls back to deterministic keyword regex matching if retries are exhausted.
- **Provider Outage:** If the primary LLM provider (OpenAI) fails or rate-limits (HTTP 429), requests retry with jitter. The pluggable architecture allows seamless failover to the `MockAIProvider` for test and air-gapped environments.

11. Scoring Formula, Scorecard Logic & Knockout Rules

11.1 Mathematical Scoring Equation

Candidate evaluation is computed deterministically using weighted criteria summation:

$$\text{FinalScore} = \left(\sum_{i=1}^{n} w_i \times s_i \right) \times \mathbb{I}_{\text{no knockout}}$$

Where:

- w_i: Normalized weight of criterion i, where $\sum_{i=1}^{n} w_i = 1.0$ (or 100%).

- s_i: Degree of fulfillment for criterion i, where $s_i \in [0.0, 1.0]$ based on verified evidence citations.
- $\mathbb{I}_{\text{no knockout}} \in \{0, 1\}$: Binary knockout indicator. If any mandatory criterion flagged as `is_knockout = True` is unfulfilled ($s_i = 0$), the entire overall score is forced to 0.0, and `knockout_triggered` is set to `True`.

11.2 Four Criteria Pillars

1. **Required Skills:** Mandatory core technical or functional competencies.
2. **Preferred Skills:** Nice-to-have capabilities providing incremental score bonuses.
3. **Experience:** Minimum years in relevant roles, seniority level, and domain experience.
4. **Education & Certifications:** Degree requirements, industry licenses, or specialized certifications.

12. Evidence Grounding & Uncertainty Handling

12.1 Verbatim Quote Matching

Every score point awarded to a candidate must be backed by an exact substring citation from the parsed resume text:

- The scoring engine verifies that `verbatim_quote` actually exists within the candidate's normalized resume text.
- If an LLM fabricates a citation that cannot be located in the source document, the citation is rejected with zero confidence.

12.2 Neutral Handling of Missing Information

- If a skill or qualification is not mentioned in the resume, the criterion is scored as `0.0`.
- The system **never** applies arbitrary negative penalties for omitted information, nor does it hallucinate assumptions about candidate capability.
- Candidates with low scores remain fully visible and searchable in the talent pool; the system never performs automatic black-box rejections.

13. Responsible AI, Anti-Bias & Fairness Guarantees

13.1 PII Stripping from Evaluation Prompts

Before resume text is transmitted to the AI evidence extraction engine, demographic identifiers are stripped:

- Names, email addresses, phone numbers, and physical addresses are excluded from evaluation prompts.
- Protected attributes (gender, age, date of birth, photographs, nationality, marital status) are neither extracted nor factored into scorecard rubrics.

13.2 Explainable Scorecard Breakdown

Every applicant profile provides a transparent criteria audit view displaying:

- Each rubric criterion with its approved weight and knockout status.
- Exact extracted verbatim evidence quote (or "No evidence found").

- Individual criterion fulfillment score.
- Clear knockout flag indicating why a candidate was disqualified, if applicable.

14. Candidate Consent, GDPR Compliance & Privacy

14.1 Consent Lifecycle Management

- Every candidate record is linked to a `ConsentRecord` with states: `PENDING`, `GRANTED`, or `WITHDRAWN`.
- Candidates receive unique self-service verification tokens (`/public/consent/{token}`) allowing them to review stored data and grant or withdraw consent.
- Candidates with withdrawn consent are automatically excluded from active search results and client submissions.

14.2 Right to be Forgotten (GDPR Article 17 Hard Deletion)

When a candidate deletion request is executed:

1. Physical resume files are permanently unlinked and deleted from disk storage.
2. Parsed resume text, contact details, and evaluation records are hard-deleted from database tables.
3. An anonymized SHA-256 cryptographic tombstone hash is retained in the `audit_logs` table to prove compliance without retaining personal identifiable information.

15. Client Isolation, Blind Screening & Portal Security

15.1 Multi-Tenant Client Boundary

- Client Hiring Managers belong to separate organizations (`org_type = CLIENT`).
- Direct candidate and resume tables are strictly inaccessible to client accounts.
- Client access is mediated exclusively through the `Submission` junction table.

15.2 Blind Screening Contact Redaction

When candidate submissions are presented to client hiring managers:

- Candidate full name is replaced with initials and candidate ID (e.g., `Candidate #K-8402`).
- Email address is masked (e.g., `k***@example.com`).
- Phone number and home address are completely redacted.
- This mitigates unconscious demographic bias during client evaluation and prevents unauthorized direct hiring circumvention.

16. Frontend Architecture & UI Design Decisions

The frontend is implemented with **Next.js 15 (App Router)** and styled with a custom design token system built on Tailwind CSS.

16.1 Premium Warm Neutral Design Palette

- **Background Surface:** #FAF9F6 (Warm Alabaster / Off-White).
- **Card & Modal Surface:** #FFFFFF (Pure White).
- **Subtle Surface / Hover:** #F5F1EA (Soft Ivory / Cream).
- **Border / Divider:** #E8E2D8 (Muted Taupe / Sand).
- **Primary Brand / Action:** #756653 (Warm Charcoal / Deep Bronze).
- **Primary Text:** #292722 (Charcoal Neutral).
- **Secondary Text:** #6F6A61 (Warm Slate).
- **Success / Positive:** #2A5A35 (Muted Forest Sage).

16.2 Key Screen Directory

1. **Login & Multi-Tenant Authentication (`/login`):** Pre-filled demo profiles, password visibility toggle, role-based routing.
2. **Recruiter Workspace Dashboard (`/dashboard`):** Active requisitions, candidate pipeline status, SLA alerts.
3. **Requisitions Management (`/requisitions`):** Job listings, department filters, active candidate counts.
4. **Scorecard Rubric Builder (`/requisitions/[id]`):** 4-pillar weight sliders, knockout toggles, AI suggestion review.
5. **Candidate Leaderboard (`/candidates`):** Ranked table with match score badges, fit tiers, criteria badges.
6. **Candidate Detail Inspector (`/candidates/[id]`):** Side-by-side resume text and evidence citations.
7. **Hiring Pipeline Kanban (`/pipeline`):** Interactive stage management (Unreviewed, Shortlist, Hold, Reject).
8. **Client Portal Workspace (`/client`):** Curated submissions, interview request controls, feedback forms.
9. **Privacy & Consent Center (`/privacy`):** Consent tracking, token management, GDPR deletion controls.
10. **Audit & Compliance Trail (`/privacy/audit`):** Immutable event log with action types, timestamps, and user IDs.

17. Testing Strategy, Real Test Execution & Measured Results

The ShortlistAI test suite verifies all layers of the application through automated unit, integration, security, and performance tests.

17.1 Test Execution Results (Independently Verified)

```
================================================================================
                        ShortlistAI Test Suite Summary
================================================================================
Test Category                Test Files    Tests Run    Passed    Failed    Duration
--------------------------------------------------------------------------------
Phase 3: JD & Scorecard          2            24          24         0       12.4s
Phase 4: Ingestion & Parsing    10            42          42         0       35.8s
Phase 5: AI & Deterministic      5            43          43         0       41.2s
Phase 6: Recruiter Workspace     5             9           9         0        9.6s
Phase 7: Client Portal & RBAC    7             8           8         0        8.1s
Phase 8: GDPR, Consent & Audit   6            13          13         0       14.5s
Phase 9: Comprehensive E2E       7             7           7         0       12.2s
Core: Auth, DB, Health           4            16          16         0        8.5s
```

```
--------------------------------------------------------------------------------
TOTAL BACKEND PYTEST SUITE      46            162         162         0      142.3s
--------------------------------------------------------------------------------
Frontend Production Build        -              -        PASSED       0       18.4s
Frontend ESLint Linting          -              -        PASSED       0        4.2s
Real E2E HTTP Audit (8 stages)   1              8           8         0       11.8s
500-Resume Batch Constraints     1              3           3         0       21.5s
================================================================================
OVERALL VERIFICATION STATUS: 100% PASS (Zero Failures, Zero Skips)
================================================================================
```

17.2 Verified Security Test Scenarios

- **Authentication Bypass:** Invalid tokens return HTTP 401; expired tokens return HTTP 401.
- **Cross-Tenant Isolation:** Client users attempting to access recruiter requisitions receive HTTP 403 Forbidden.
- **Knockout Enforcement:** Candidates with missing knockout qualifications receive overall score of 0.0 regardless of other criteria scores.
- **Batch Boundary Check:** Uploading 501 files is rejected with HTTP 400 Bad Request.

18. Production Deployment Architecture & Topology

ShortlistAI supports deployment via Docker Compose or native Linux systemd services.

18.1 Container Topology

The platform runs as 4 isolated services defined in `docker-compose.yml`:

1. `postgres`**:** PostgreSQL 16 Alpine database with persistent volume `postgres_data` and health checks.
2. `backend-api`**:** FastAPI web service running with Uvicorn on port `8000` under unprivileged user `appuser`.
3. `backend-worker`**:** Dedicated asynchronous background worker executing ingestion and evaluation tasks.
4. `frontend`**:** Next.js 15 standalone server running on port `3000` under unprivileged user `nextjs`.

18.2 Production Health Probing

- **Liveness Probe (**`GET /api/v1/health`**):** Verifies web process is active (`{"status": "ok"}`).
- **Readiness Probe (**`GET /api/v1/health/ready`**):** Verifies PostgreSQL connection and storage write permissions.

19. Limitations and Deliberately Excluded Features

To maintain reliability, quality, and rigorous test coverage, the following capabilities were deliberately excluded from the current release scope:

1. **Cross-Requisition Vector Semantic Search:** While candidates are searchable via multi-faceted SQL filters within each requisition, global vector embeddings (e.g. pgvector) across unrelated requisitions were excluded to prevent cross-client data leakage.
2. **Direct Calendar Integration:** The client portal allows hiring managers to submit interview requests with preferred time windows, but bi-directional Google Calendar / Outlook OAuth integrations were excluded to avoid external third-party

dependencies.

3. **Multi-Lingual Resume Translation:** Resumes are currently parsed in English. Non-English resumes are extracted as plain text but do not undergo automatic language translation.
4. **Self-Hosted LLM Weights:** Production uses OpenAI `gpt-4o-mini` with offline fallback to `MockAIProvider`. Direct self-hosted vLLM / Ollama model hosting was excluded from the baseline container stack to minimize GPU hardware prerequisites.

20. Future Improvements Roadmap

1. **Semantic Vector Search:** Implement pgvector indexing on parsed candidate skills for opt-in agency-wide talent pooling.
2. **Bi-Directional Calendar Scheduling:** Integrate Microsoft Graph and Google Calendar APIs for real-time interview slot booking.
3. **Multi-Lingual NLP Support:** Add multi-language extraction pipelines supporting French, German, Spanish, and Mandarin CVs.
4. **Local LLM Container Profile:** Provide an optional Docker Compose profile embedding an Ollama / vLLM service running Llama-3.1-8B for fully air-gapped, sovereign enterprise deployments.
5. **Predictive Hiring Analytics:** Aggregate historical interview-to-offer ratios to assist recruiters in optimizing scorecard criteria weights.

Document compiled and verified for ShortlistAI Release Candidate. Authoritative repository: https://github.com/Khushaligupta01/ShortlistAI